

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester VI Examination, 2013-14

Paper No. STA/STB-601: Numerical Analysis

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

Question 1.

(a) Prove the following identities (symbols have their usual meaning):

(i) $\Delta \cdot E^{-1} = \nabla$

(ii) $(\Delta - \nabla)(\Delta + \nabla) = \Delta^2 - \nabla^2$

[7]

(b) State and prove the fundamental theorem of difference calculus.

[7]

Question 2.(a) Show that $\Delta^n(x^{(n)}) = n!$ and hence find the value of $\Delta^6(x^6)$.

[7]

(b) Obtain the missing term y_2 in the following table:

x	0	1	2	3	4
y	1	3	–	13	21

[7]

Question 3.

(a) State and prove the Newton–Gregory forward interpolation formula.

[7]

(b) Using Stirling's interpolation formula, find the value of y_{35} given that $y_{20} = 512$, $y_{30} = 439$, $y_{40} = 346$, $y_{50} = 243$.

[7]

Question 4.

(a) State and prove Newton's divided difference interpolation formula for unequal intervals.

[7]

(b) Using the Lagrange interpolation formula, find the polynomial form of the function $f(x)$ tabulated below:

x	0	1	4	5
$f(x)$	4	3	24	39

[7]

Question 5.

(a) What is inverse interpolation? Tabulate $y = x^{1/3}$ for $x = 2, 3, 4, 5$ and calculate the cube root of 10 correct to three decimal places. [7]

(b) From the following table, compute $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x = 1$:

x	1.0	1.5	2.0	2.5	3.0
y	2.7183	4.4817	7.3891	12.182	20.086

[7]

Question 6.

(a) State and prove Simpson's three-eighths ($3/8$) rule of numerical integration. [7]

(b) Calculate the value of π from $\int_0^1 \frac{dx}{1+x^2}$, taking six equal intervals and using Simpson's one-third rule. [7]

Question 7.

(a) Sum the following series to the first n terms:

$$1, 12, 36, 80, 150, 252, \dots$$

[7]

(b) Find the sum of the first n terms of a series whose k -th term T_k is given by $T_k = 2^k(3+k)$. [7]

Question 8.

(a) Using the Runge-Kutta method, approximate y when $x = 0.2$, given that $y = 1$ when $x = 0$ and $\frac{dy}{dx} = x + y$. [7]

(b) Solve $\frac{dy}{dx} = 1 - y$, $y(0) = 0$, in the range $0 \leq x \leq 0.3$ using the modified Euler's method with step size $h = 0.1$. [7]

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